

Engineering Mathematics - Complete Notes

Coal India Limited - Management Trainee (Systems) | CBT Preparation

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Engineering Mathematics - CIL MT Systems Complete Notes

- This guide covers the COMPLETE Engineering Mathematics syllabus for the **Coal India Limited (CIL) Management Trainee** Systems discipline objective CBT.
- Difficulty target: **conceptual-to-moderate** (direct formula application), but depth is kept at **GATE-level** for safety.
- Five pillars: **Discrete Mathematics, Linear Algebra, Calculus, Probability & Statistics, and Numerical Methods.**
- How to use: read the strict definition, memorize the red formulas, then absorb each **CBT Trap** which names the exact confusions examiners exploit.

1. Mathematical Logic

1.1 Propositional Logic and Connectives

- A **proposition** is a declarative statement that is either **true (T)** or **false (F)**, but never both.
- A **connective** combines propositions into compound propositions; the five standard ones are below.
- **Negation** (NOT, symbol $\neg p$ or $\sim p$): reverses truth value.
- **Conjunction** (AND, symbol $p \wedge q$): true only when **both** are true.
- **Disjunction** (OR, symbol $p \vee q$): true when **at least one** is true (inclusive OR).
- **Implication** (IF-THEN, symbol $p \rightarrow q$): false **ONLY** when **p is true and q is false**.
- **Biconditional** (IFF, symbol $p \leftrightarrow q$): true when **p and q have the same truth value**.

Connective	Symbol	True when
NOT	$\neg p$	p is false
AND	$p \wedge q$	both true
OR	$p \vee q$	at least one true
Implication	$p \rightarrow q$	except T→F
Biconditional	$p \leftrightarrow q$	same values

- For **n** propositional variables a truth table has **2^n** rows.
- Implication equivalents: $p \rightarrow q \equiv \neg p \vee q$.
- Contrapositive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$ (logically equivalent).
- Converse of $p \rightarrow q$ is $q \rightarrow p$ (NOT equivalent).
- Inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$ (NOT equivalent, but equivalent to converse).

Memory Trick

- "Implication is only a **liar when premise is true but conclusion false**" — remember the single F row of $p \rightarrow q$.

CBT Trap

- **Converse and inverse are NOT equivalent to the original**; only the **contrapositive** is.
- OR in logic is **inclusive** by default; XOR (**exclusive or**) is true when **exactly one** operand is true.
- A vacuous implication with a **false premise** is always **true**.

1.2 Truth Tables; Tautology, Contradiction, Contingency

- A **tautology** is a compound proposition that is **always true** for every assignment (e.g., $p \vee \neg p$).
- A **contradiction** is **always false** (e.g., $p \wedge \neg p$).
- A **contingency** is neither — its value **depends on inputs**.

- $p \rightarrow q$ is a tautology iff p logically implies q .

CBT Trap

- **Tautology vs satisfiable**: a tautology is true in ALL rows; satisfiable needs only **at least one** true row.

1.3 Logical Equivalences; De Morgan's Laws

- Two propositions are **logically equivalent** ($\equiv$) when they have **identical truth tables**.
- De Morgan: $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$.
- Identity: $p \wedge T \equiv p$, $p \vee F \equiv p$.
- Domination: $p \vee T \equiv T$, $p \wedge F \equiv F$.
- Idempotent: $p \vee p \equiv p$, $p \wedge p \equiv p$.
- Distributive: $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
- Absorption: $p \vee (p \wedge q) \equiv p$.

Law	Form
Commutative	$p \wedge q \equiv q \wedge p$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

CBT Trap

- When negating, De Morgan **flips AND to OR** and vice versa AND negates each term — students forget to flip the connective.

1.4 Normal Forms: CNF and DNF

- **DNF (Disjunctive Normal Form)**: an **OR of ANDs** — disjunction of conjunctive clauses (minterms).
- **CNF (Conjunctive Normal Form)**: an **AND of ORs** — conjunction of disjunctive clauses (maxterms).
- Every proposition can be written in both **CNF** and **DNF**.

Memory Trick

- **DNF = D** for "Disjunction outside"; **CNF = C** for "Conjunction outside".

CBT Trap

- Do not confuse **CNF (product of sums)** with **DNF (sum of products)**; SOP = DNF, POS = CNF.

1.5 Rules of Inference; Validity of Arguments

- **Modus Ponens**: from p and $p \rightarrow q$ infer q .
- **Modus Tollens**: from $\neg q$ and $p \rightarrow q$ infer $\neg p$.
- **Hypothetical Syllogism**: from $p \rightarrow q$ and $q \rightarrow r$ infer $p \rightarrow r$.
- **Disjunctive Syllogism**: from $p \vee q$ and $\neg p$ infer q .
- **Addition**: from p infer $p \vee q$.
- **Simplification**: from $p \wedge q$ infer p .
- An argument is **valid** when the conclusion is true whenever all premises are true.

CBT Trap

- **Affirming the consequent** (from q and $p \rightarrow q$ conclude p) and **denying the antecedent** are **fallacies**, not valid rules.

1.6 First-Order (Predicate) Logic: Predicates and Quantifiers

- A **predicate** $P(x)$ becomes a proposition once x is assigned a value or quantified.
- **Universal quantifier** $\forall x P(x)$: "for all x , $P(x)$ " — true if P holds for **every** element.
- **Existential quantifier** $\exists x P(x)$: "there exists x " — true if P holds for **at least one** element.

1.7 Nested Quantifiers and Negation of Quantified Statements

- Negation rule: $\neg \forall x P(x) \equiv \exists x \neg P(x)$ and $\neg \exists x P(x) \equiv \forall x \neg P(x)$.
- $\forall x \exists y$ differs from $\exists y \forall x$: order of quantifiers changes meaning.

CBT Trap

- $\forall x \exists y P(x,y)$ (each x has some y) is **weaker** than $\exists y \forall x P(x,y)$ (one y works for all x); swapping order is a classic trap.
- To negate, **swap** $\forall \leftrightarrow \exists$ and negate the inner predicate.

2. Set Theory and Relations

2.1 Sets, Subsets, Power Set, Operations

- A **set** is an unordered collection of distinct objects (elements).
- $A \subseteq B$ (subset): every element of A is in B; **proper subset** $A \subset B$ excludes equality.
- Power set** $P(A)$: set of all subsets; for $|A| = n$, $|P(A)| = 2^n$.
- Union** $A \cup B$: elements in A or B.
- Intersection** $A \cap B$: elements in both.
- Difference** $A - B$: in A but not B.
- Complement** A' : elements in universal set U not in A.

Operation	Notation	Definition
Union	$A \cup B$	in A or B
Intersection	$A \cap B$	in both
Difference	$A - B$	in A not B
Symmetric diff	$A \oplus B$	in exactly one

- Number of subsets of size r: $C(n, r)$; total subsets 2^n ; proper subsets $2^n - 1$.

CBT Trap

- Empty set** $\emptyset$ is a subset of EVERY set; $\emptyset \neq \{\emptyset\}$ ($|\{\emptyset\}| = 1$).
- Power set of empty set has **1** element, not 0.

2.2 Venn Diagrams; Inclusion-Exclusion Principle

- Two-set: $|A \cup B| = |A| + |B| - |A \cap B|$.
- Three-set: $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.

Memory Trick

- Inclusion-exclusion alternates signs: **add singles, subtract pairs, add triples**.

2.3 Cartesian Product

- $A \times B$ = set of ordered pairs (a, b); $|A \times B| = |A| \times |B|$.
- $A \times B \neq B \times A$ in general (order matters).

2.4 Relations: Properties

- A **relation** R on set A is a subset of $A \times A$.
- Reflexive**: $(a, a) \in R$ for all a.
- Symmetric**: $(a, b) \in R \Rightarrow (b, a) \in R$.
- Antisymmetric**: $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$.
- Transitive**: $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$.
- On a set of n elements there are $2^{(n^2)}$ relations.

Property	Condition
Reflexive	$(a, a) \in R \forall a$
Symmetric	$aRb \Rightarrow bRa$

Property	Condition
Antisymmetric	$aRb \wedge bRa \Rightarrow a=b$
Transitive	$aRb \wedge bRc \Rightarrow aRc$

CBT Trap

- **Antisymmetric is NOT "not symmetric"**; a relation can be both symmetric and antisymmetric (only the diagonal).
- Number of **reflexive** relations = 2^{n^2-n} ; symmetric = $2^{n(n+1)/2}$.

2.5 Equivalence Relations and Equivalence Classes

- An **equivalence relation** is **reflexive + symmetric + transitive** (RST).
- It **partitions** the set into disjoint **equivalence classes**.
- Number of equivalence relations on n elements = **Bell number B(n)**.

CBT Trap

- A **partition** corresponds one-to-one with an **equivalence relation**; partition blocks are the classes.

2.6 Partial Order Relations; Hasse Diagrams

- A **partial order** (POSET relation) is **reflexive + antisymmetric + transitive**.
- A **Hasse diagram** draws the partial order with **transitive and reflexive edges removed** and direction implied upward.

CBT Trap

- **Equivalence vs partial order**: equivalence is symmetric; partial order is **antisymmetric** — opposite middle property.

2.7 Functions: Injective, Surjective, Bijective

- **Injective (one-one)**: distinct inputs map to distinct outputs.
- **Surjective (onto)**: every codomain element is hit.
- **Bijective**: both injective and surjective; has an **inverse**.
- Composition **$(f \circ g)(x) = f(g(x))$** .
- From a set of size m to size n: total functions n^m ; injective functions $\frac{n!}{(n-m)!}$ ($m \leq n$); surjective via inclusion-exclusion.

CBT Trap

- **Onto requires codomain = range**; counting onto functions uses **inclusion-exclusion**, not a simple factorial.
- **$(f \circ g)$** means apply **g first**, then f.

2.8 Countable and Uncountable Sets

- A set is **countable** if it is finite or has a bijection with $\mathbb{N}$ (natural numbers).
- $\mathbb{N}$ and $\mathbb{Z}$ are countable; $\mathbb{R}$ is **uncountable** (Cantor's diagonal argument).

CBT Trap

- **Rationals are countable** despite being dense; **reals are uncountable**.

3. Partial Orders and Lattices

3.1 POSET; Upper/Lower Bounds, LUB, GLB

- A **POSET** is a set with a partial order (**reflexive, antisymmetric, transitive**).
- **Upper bound** of subset S: an element $\geq$ every element of S.
- **LUB (supremum/join)**: the **least** upper bound.
- **GLB (infimum/meet)**: the **greatest** lower bound.

3.2 Lattices; Bounded, Distributive, Complemented

- A **lattice** is a POSET in which **every pair** of elements has both a **LUB (join, $\vee$)** and **GLB (meet, $\wedge$)**.
- **Bounded lattice**: has a **greatest element 1** and **least element 0**.
- **Distributive lattice**: meet distributes over join and vice versa.
- **Complemented lattice**: every element has a complement.

Lattice type	Key property
Bounded	has 0 and 1
Distributive	$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
Complemented	every element has complement
Boolean algebra	bounded + distributive + complemented

CBT Trap

- Every **chain (totally ordered set)** is a **distributive lattice**.
- A **Boolean algebra** is a **complemented distributive** lattice.

3.3 Hasse Diagram Interpretation

- Top node = **greatest element**; bottom = **least element**.
- An element covers another if directly connected with no intermediate.

CBT Trap

- Not every POSET is a lattice; check that **both LUB and GLB exist for all pairs**.

4. Algebraic Structures

4.1 Binary Operations and Properties

- A **binary operation** $*$ on set S maps $S \times S \rightarrow S$.
- **Closure**: $a*b \in S$ for all a, b .
- **Associativity**: $(a*b)*c = a*(b*c)$.
- **Identity e** : $a*e = e*a = a$.
- **Inverse**: $a*a^{-1} = e$.

4.2 Semigroup and Monoid

- **Semigroup** = closure + associativity.
- **Monoid** = semigroup + **identity element**.

4.3 Groups; Abelian Groups; Subgroups

- **Group** = monoid + **inverse** for every element (closure, associativity, identity, inverse).
- **Abelian (commutative) group**: additionally $a*b = b*a$.
- **Subgroup**: a subset that is itself a group under the same operation.

Structure	Properties
Semigroup	closure + associative
Monoid	+ identity
Group	+ inverse
Abelian group	+ commutative

Memory Trick

- Climb the ladder: **Semigroup** $\rightarrow$ **Monoid** $\rightarrow$ **Group** $\rightarrow$ **Abelian** adds **associativity-already, identity, inverse, commutativity**.

4.4 Order of a Group/Element; Cyclic Groups

- **Order of a group** = number of elements; **order of an element a** = smallest k with $a^k = e$.
- A **cyclic group** is generated by a single element; every cyclic group is **abelian**.
- **Lagrange's theorem**: order of a subgroup **divides** the order of the group.

CBT Trap

- **Group needs ALL four** axioms; missing inverse $\rightarrow$ only a monoid. Commutativity is NOT required for a group.

5. Graph Theory

5.1 Graph Terminology

- A **graph** $G = (V, E)$ has a vertex set V and edge set E .
- **Degree** of a vertex = number of incident edges.
- A **path** is a walk with no repeated vertices; a **cycle** returns to start.

5.2 Handshaking Lemma

- **Sum of degrees = 2 × number of edges:** $\sum \deg(v) = 2|E|$.
- The number of **odd-degree vertices** is always **even**.

CBT Trap

- Each edge contributes **2** to the degree sum; a common error is forgetting the factor of 2.

5.3 Types: Simple, Complete, Bipartite, Regular, Planar

- **Simple graph:** no loops or multiple edges.
- **Complete graph K_n :** every pair connected; edges $n(n-1)/2$.
- **Bipartite:** vertices split into two sets with edges only across; **no odd cycles**.
- **Regular:** every vertex has equal degree.
- **Complete bipartite $K(m,n)$:** edges $m \times n$.

Graph	Edges
Complete K_n	$n(n-1)/2$
Complete bipartite $K_{m,n}$	$m \cdot n$
Tree on n vertices	$n-1$
Cycle C_n	n

CBT Trap

- A graph is **bipartite iff it has NO odd-length cycle**.

5.4 Connectivity; Components

- A graph is **connected** if there is a path between every pair.
- A **connected component** is a maximal connected subgraph.

5.5 Eulerian and Hamiltonian Graphs

- **Eulerian circuit** (uses every edge once, returns to start) exists iff connected and **every vertex has even degree**.
- **Eulerian path** (not closed) exists iff **exactly 0 or 2 vertices have odd degree**.
- **Hamiltonian cycle** visits every **vertex** exactly once; **no simple necessary-and-sufficient condition** (NP-hard).

Concept	Touches	Condition
Eulerian	every edge	even degrees
Hamiltonian	every vertex	no simple condition

CBT Trap

- **Euler = edges, Hamilton = vertices**; Euler has an easy degree condition, Hamilton does NOT.

5.6 Trees and Spanning Trees

- A **tree** is a connected acyclic graph with **$n-1$** edges on n vertices.
- A tree has **exactly one path** between any two vertices.
- A **spanning tree** of a connected graph includes all vertices with **$n-1$** edges.
- Number of spanning trees of **$K_n = n^{n-2}$** (Cayley's formula).

CBT Trap

- Adding any edge to a tree creates **exactly one cycle**; removing any edge **disconnects** it.

5.7 Graph Coloring; Chromatic Number

- **Chromatic number $\chi(G)$** = minimum colors to color vertices so adjacent ones differ.
- **$\chi(K_n) = n$; $\chi(\text{bipartite}) = 2$** (if it has edges); **$\chi(\text{cycle } C_n) = 2$** if n even, **3** if n odd.
- **Four Color Theorem**: every planar graph is **4-colorable**.

CBT Trap

- **$\chi(\text{odd cycle}) = 3$** , not 2; bipartite needs exactly 2.

5.8 Matching

- A **matching** is a set of edges with no shared vertices.
- A **perfect matching** covers every vertex.

5.9 Planar Graphs; Euler's Formula

- A **planar graph** can be drawn with no edge crossings.
- **Euler's formula: $V - E + F = 2$** for connected planar graphs.
- Necessary condition: **$E \leq 3V - 6$** (simple, $V \geq 3$); for bipartite **$E \leq 2V - 4$** .
- **K_5 and $K_{3,3}$** are the basic **non-planar** graphs (Kuratowski).

CBT Trap

- **Euler's formula $V - E + F = 2$** counts the **outer (unbounded) face** as one of the F faces.

6. Combinatorics

6.1 Counting Principles

- **Sum rule:** if tasks are mutually exclusive, total ways = $n_1 + n_2$.
- **Product rule:** for sequential independent choices, total = $n_1 \times n_2$.

6.2 Permutations and Combinations

- **Permutation** $P(n,r) = n!/(n-r)!$ — order matters.
- **Combination** $C(n,r) = n!/(r!(n-r)!)$ — order does not matter.
- $C(n,r) = C(n, n-r)$; circular permutations of $n = (n-1)!$.
- Permutations with repetition of n items = n^r .

Concept	Formula
Permutation	$n!/(n-r)!$
Combination	$n!/(r!(n-r)!)$
Circular perm	$(n-1)!$
With repetition	n^r

CBT Trap

- **Order matters** → **permutation**; **order does not** → **combination**. Mixing these is the #1 counting error.

6.3 Pigeonhole Principle

- If n items are placed in k boxes and $n > k$, some box has ≥ 2 items.
- Generalized: some box has $\geq \lceil n/k \rceil$ items.

CBT Trap

- Use **ceiling** $\lceil n/k \rceil$, not floor, for the guaranteed minimum in some box.

6.4 Inclusion-Exclusion (Counting)

- $|A \cup B| = |A| + |B| - |A \cap B|$ extends to derangements and onto-function counts.
- Number of **derangements** $D(n) = n! \sum (-1)^k / k!$.

6.5 Binomial Theorem and Coefficients

- $(x + y)^n = \sum C(n,k) x^{n-k} y^k$.
- Sum of coefficients = 2^n (set $x=y=1$); $\sum (-1)^k C(n,k) = 0$.
- General term $T(k+1) = C(n,k) x^{n-k} y^k$.

CBT Trap

- The **(k+1)-th term** uses $C(n, k)$; off-by-one indexing is a common error.

6.6 Recurrence Relations

6.6.1 Forming Recurrences

- Express a_n in terms of earlier terms, e.g., $a_n = a_{(n-1)} + a_{(n-2)}$ (Fibonacci).

6.6.2 Solving Linear Homogeneous Recurrences

- For $a_n = c_1 a_{(n-1)} + c_2 a_{(n-2)}$, form **characteristic equation** $x^2 = c_1 x + c_2$.
- Distinct roots r_1, r_2 : $a_n = A r_1^n + B r_2^n$.
- Repeated root r : $a_n = (A + Bn) r^n$.

6.6.3 Particular Solutions

- For non-homogeneous, total solution = **homogeneous + particular**.

Root case	General solution
Distinct r_1, r_2	$A r_1^n + B r_2^n$
Repeated r	$(A+Bn) r^n$
Complex	trigonometric form

CBT Trap

- Repeated roots** require the extra factor n ($(A+Bn)r^n$); forgetting it is a classic mistake.

6.7 Generating Functions

- A **generating function** encodes a sequence as coefficients of a power series $\sum a_n x^n$.
- $1/(1-x) = \sum x^n$ is the basic generating function.

CBT Trap

- Generating functions turn **recurrences** into **algebraic equations** in the series variable.

7. Matrices and Determinants

7.1 Types of Matrices

- **Diagonal matrix:** non-zero entries only on the main diagonal.
- **Identity matrix I :** diagonal of **1s**; $AI = IA = A$.
- **Symmetric matrix:** $A = A^T$ ($a_{ij} = a_{ji}$).
- **Skew-symmetric:** $A = -A^T$; diagonal entries are **0**.
- **Orthogonal matrix:** $AA^T = I$, so $A^{-1} = A^T$; determinant ± 1 .

Type	Defining property
Symmetric	$A = A^T$
Skew-symmetric	$A = -A^T$, diag 0
Orthogonal	$AA^T = I$
Identity	diag of 1s

CBT Trap

- **Skew-symmetric** forces **zero diagonal**; an odd-order skew-symmetric matrix has determinant **0**.

7.2 Matrix Operations

- Addition is **element-wise** and requires same dimensions.
- Multiplication: $(AB)_{ij} = \sum a_{ik} b_{kj}$; defined when **columns of A = rows of B** .
- Matrix multiplication is **associative** and **distributive** but **NOT commutative**.
- $(AB)^T = B^T A^T$; $(AB)^{-1} = B^{-1} A^{-1}$ (reverse order).
- Naive multiplication of $n \times n$ matrices is $O(n^3)$; **Strassen** is $O(n^{2.81})$.

CBT Trap

- $AB \neq BA$ in general; transpose and inverse of a product **reverse the order**.

7.3 Determinant: Properties and Computation

- $\det(I) = 1$; $\det(A^T) = \det(A)$.
- $\det(AB) = \det(A) \cdot \det(B)$.
- $\det(kA) = k^n \det(A)$ for $n \times n$.
- Swapping two rows **multiplies det by -1** ; identical rows give **det = 0**.
- $\det(A^{-1}) = 1/\det(A)$.
- 2×2 : $\det = ad - bc$.

CBT Trap

- $\det(kA) = k^n \det(A)$, NOT $k \cdot \det(A)$ — the scalar is raised to the order n .
- $\det(A+B) \neq \det(A) + \det(B)$ in general.

7.4 Inverse of a Matrix; Adjoint

- $A^{-1} = \text{adj}(A) / \det(A)$, exists iff $\det(A) \neq 0$ (non-singular).
- $\text{adj}(A)$ = **transpose of the cofactor matrix**.
- $A \cdot \text{adj}(A) = \det(A) \cdot I$.

- $\det(\text{adj } A) = \det(A)^{(n-1)}$.

CBT Trap

- A matrix is invertible iff $\det \neq 0$ iff **full rank** iff **0 is not an eigenvalue**.

7.5 Rank of a Matrix

- **Rank** = number of **linearly independent rows (= columns)** = order of the largest non-zero minor.
- Rank equals the number of **non-zero rows in row-echelon form**.
- For $n \times n$: **full rank $n \Leftrightarrow$ non-singular**.

CBT Trap

- Rank of a **non-zero matrix** is at least **1**; rank 0 only for the **zero matrix**.

7.6 Elementary Row Operations

- Three types: **row swap**, **scaling a row**, **adding a multiple of one row to another**.
- They preserve **rank** and the **solution set** of a system (but can change determinant).

8. System of Linear Equations

8.1 Homogeneous vs Non-Homogeneous

- **Homogeneous:** $Ax = 0$; always has the **trivial solution** $x = 0$.
- **Non-homogeneous:** $Ax = b$, $b \neq 0$.

8.2 Consistency and Number of Solutions (Rank Method)

- Let $r = \text{rank}(A)$, $r' = \text{rank}([A|b])$, n = number of unknowns.
- **Inconsistent** (no solution) if $r < r'$.
- **Unique solution** if $r = r' = n$.
- **Infinite solutions** if $r = r' < n$ ($n - r$ free variables).

Condition	Solutions
$r < r'$	none (inconsistent)
$r = r' = n$	unique
$r = r' < n$	infinite

- Homogeneous $Ax=0$ has non-trivial solutions iff **rank** $< n$ iff **det(A) = 0**.

CBT Trap

- Always compare rank of **A** vs the **augmented matrix** $[A|b]$; equality decides consistency.

8.3 Gaussian Elimination

- Reduce $[A|b]$ to **upper triangular (row-echelon)** form, then **back-substitute**.
- Complexity **$O(n^3)$** .

8.4 Cramer's Rule

- For $Ax = b$ with $\det(A) \neq 0$: **$x_i = \det(A_i)/\det(A)$** , where A_i replaces column i with b .
- Practical only for **small systems** (2×2 , 3×3).

CBT Trap

- **Cramer's rule fails when $\det(A) = 0$** ; it cannot detect infinite vs no solution.

9. Eigenvalues and Eigenvectors

9.1 Characteristic Equation

- $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ where λ is an **eigenvalue**, $\mathbf{v} \neq \mathbf{0}$ the **eigenvector**.
- Characteristic equation: $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$.

9.2 Computation

- Solve the **characteristic polynomial** for λ , then solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ for each eigenvector.

9.3 Properties

- **Sum of eigenvalues = trace(A)** (sum of diagonal).
- **Product of eigenvalues = det(A)**.
- Eigenvalues of a **triangular matrix** are its **diagonal entries**.
- **Symmetric matrices** have **real** eigenvalues; orthogonal eigenvectors.
- Eigenvalues of $\mathbf{A}^{-1}$ are $1/\lambda$; of $\mathbf{A}^k$ are λ^k .

Property	Result
Sum of eigenvalues	trace
Product	determinant
Triangular matrix	diagonal entries
$\mathbf{A}^{-1}$ eigenvalues	$1/\lambda$

CBT Trap

- **Trace = sum, det = product** of eigenvalues — fast checks. A singular matrix has **0** as an eigenvalue.

9.4 Cayley-Hamilton Theorem

- Every square matrix **satisfies its own characteristic equation**.
- Used to compute $\mathbf{A}^{-1}$ and high powers of $\mathbf{A}$ without direct inversion.

9.5 Diagonalization

- $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ where $\mathbf{D}$ is diagonal of eigenvalues, $\mathbf{P}$ columns are eigenvectors.
- $\mathbf{A}$ is diagonalizable iff it has **n linearly independent eigenvectors**.

CBT Trap

- A matrix with **repeated eigenvalues** may NOT be diagonalizable if it lacks enough independent eigenvectors.

10. LU Decomposition

10.1 LU Factorization Concept and Method

- **$A = LU$** : L is **lower triangular**, U is **upper triangular**.
- **Doolittle**: L has unit diagonal; **Crout**: U has unit diagonal.
- Exists when all **leading principal minors are non-zero** (else needs pivoting $\rightarrow PA = LU$).

10.2 Solving Systems Using LU

- Solve **$Ly = b$** by **forward substitution**, then **$Ux = y$** by **back substitution**.
- Efficient for **multiple right-hand sides** since L,U are reused.
- Factorization cost **$O(n^3)$** ; each solve **$O(n^2)$** .

CBT Trap

- LU is preferred over re-inverting A for **repeated b vectors**; factor once, solve many.

11. Limits, Continuity, Differentiability

11.1 Limits and Standard Limit Results

- $\lim (\sin x)/x = 1$ as $x \rightarrow 0$.
- $\lim (1 - \cos x)/x^2 = 1/2$ as $x \rightarrow 0$.
- $\lim (e^x - 1)/x = 1$ as $x \rightarrow 0$.
- $\lim (1 + 1/n)^n = e$ as $n \rightarrow \infty$.
- L'Hopital's rule: for $0/0$ or ∞/∞ , $\lim f/g = \lim f'/g'$.

CBT Trap

- L'Hopital applies only to **indeterminate forms $0/0$ or ∞/∞** ; applying it elsewhere gives wrong answers.

11.2 Continuity

- f is **continuous at a** if $\lim_{x \rightarrow a} f(x) = f(a)$ (limit exists and equals the value).
- Continuous on an interval if continuous at every point.

11.3 Differentiability; Relation to Continuity

- f is **differentiable** at a if $f'(a)$ exists.
- **Differentiable $\Rightarrow$ continuous**, but **continuous $\nRightarrow$ differentiable** (e.g., $|x|$ at 0).

CBT Trap

- $|x|$ is continuous everywhere but **not differentiable at $x = 0$** (corner point).

11.4 Derivative Rules

- **Product rule:** $(uv)' = u'v + uv'$.
- **Quotient rule:** $(u/v)' = (u'v - uv')/v^2$.
- **Chain rule:** $d/dx f(g(x)) = f'(g(x)) \cdot g'(x)$.
- $d/dx x^n = n x^{n-1}$; $d/dx e^x = e^x$; $d/dx \ln x = 1/x$; $d/dx \sin x = \cos x$.

CBT Trap

- Quotient rule numerator is **$u'v - uv'$** (order matters); reversing the sign is a frequent slip.

12. Applications of Derivatives

12.1 Maxima and Minima

- **Critical points**: where $f'(x) = 0$ or f' undefined.
- **Local max**: f' changes + to -; **local min**: - to +.
- **Global extrema** on $[a,b]$: check critical points AND **endpoints**.

12.2 Critical Points; First and Second Derivative Tests

- **First derivative test**: sign change of f' .
- **Second derivative test**: $f'' > 0 \Rightarrow$ minimum, $f'' < 0 \Rightarrow$ maximum, $f'' = 0 \Rightarrow$ inconclusive.
- **Inflection point**: where $f'' = 0$ and concavity changes.

Test	Max	Min
f' sign	+ to -	- to +
f'' value	$f'' < 0$	$f'' > 0$

CBT Trap

- $f'' = 0$ is **inconclusive**; must fall back to the first-derivative test.

12.3 Mean Value Theorems

- **Rolle's theorem**: if f continuous on $[a,b]$, differentiable on (a,b) , and $f(a)=f(b)$, then some c has $f'(c)=0$.
- **Lagrange's MVT**: some c has $f'(c) = (f(b)-f(a))/(b-a)$.

CBT Trap

- **Rolle** is a special case of **Lagrange** when $f(a)=f(b)$; both need continuity on closed and differentiability on open interval.

13. Integration

13.1 Indefinite Integrals; Standard Integrals

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ ($n \neq -1$).
- $\int (1/x) dx = \ln|x| + C$.
- $\int e^x dx = e^x + C$; $\int \sin x dx = -\cos x + C$; $\int \cos x dx = \sin x + C$.

13.2 Methods: Substitution, By Parts, Partial Fractions

- **Integration by parts:** $\int u dv = uv - \int v du$ (choose u by **ILATE**).
- **Substitution:** change variable to simplify.
- **Partial fractions:** decompose rational functions before integrating.

Memory Trick

- **ILATE:** Inverse, Logarithmic, Algebraic, Trigonometric, Exponential — order to pick u in by-parts.

13.3 Definite Integrals and Properties

- $\int_a^b f dx = F(b) - F(a)$ (Fundamental Theorem).
- $\int_a^b f = -\int_b^a f$; $\int_a^a f = 0$.
- $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ (king property).

13.4 Area Under a Curve

- Area between curve and x-axis = $\int_a^b |f(x)| dx$.

CBT Trap

- For signed regions use $|f(x)|$ or split at zeros; otherwise positive and negative areas cancel.

14. Probability

14.1 Sample Space, Events; Axioms

- **Sample space S**: set of all outcomes; an **event** is a subset of S.
- Axioms: $0 \leq P(A) \leq 1$; $P(S) = 1$; for disjoint events $P(A \cup B) = P(A) + P(B)$.
- General addition: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

14.2 Conditional Probability; Multiplication Rule

- $P(A|B) = P(A \cap B)/P(B)$, $P(B) > 0$.
- Multiplication: $P(A \cap B) = P(A|B) \cdot P(B)$.

14.3 Independent Events

- A, B independent iff $P(A \cap B) = P(A) \cdot P(B)$, equivalently $P(A|B) = P(A)$.

CBT Trap

- **Independent $\neq$ mutually exclusive**; mutually exclusive events with non-zero probability are **dependent** ($P(A \cap B) = 0 \neq P(A)P(B)$).

14.4 Bayes' Theorem and Total Probability

- **Total probability**: $P(A) = \sum P(A|B_i) \cdot P(B_i)$.
- **Bayes**: $P(B_i|A) = P(A|B_i)P(B_i) / \sum P(A|B_j)P(B_j)$.

CBT Trap

- Bayes **reverses the conditioning**; the denominator is the **total probability** of the evidence.

14.5 Random Variables: Discrete and Continuous

- **Discrete RV**: countable values, described by a **PMF**.
- **Continuous RV**: uncountable values, described by a **PDF**.

14.6 PMF/PDF; CDF

- **PMF**: $P(X = x)$, sums to 1.
- **PDF** $f(x)$: $\int f(x) dx = 1$; $P(X=x) = 0$ for continuous.
- **CDF** $F(x) = P(X \leq x)$; non-decreasing from 0 to 1.

CBT Trap

- For a **continuous** RV, $P(X = a) = 0$; only intervals carry probability.

14.7 Expectation, Variance, Standard Deviation

- $E[X] = \sum x \cdot P(x)$ (discrete) or $\int x f(x) dx$ (continuous).
- $\text{Var}(X) = E[X^2] - (E[X])^2$.
- $\text{SD} = \sqrt{\text{Var}}$.
- $E[aX+b] = aE[X]+b$; $\text{Var}(aX+b) = a^2\text{Var}(X)$.

- $E[X+Y] = E[X]+E[Y]$ always; $\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)$ only if **independent**.

CBT Trap

- $\text{Var}(aX+b) = a^2\text{Var}(X)$ — the constant b drops out and a is **squared**.

15. Probability Distributions

15.1 Uniform Distribution

- Discrete uniform on n values: each probability $1/n$; mean $(a+b)/2$.
- Continuous uniform on $[a,b]$: $f = 1/(b-a)$, mean $(a+b)/2$, variance $(b-a)^2/12$.

15.2 Binomial Distribution

- $P(X=k) = C(n,k) p^k (1-p)^{n-k}$.
- Mean np ; variance $np(1-p)$.
- Models n independent trials with two outcomes.

15.3 Poisson Distribution

- $P(X=k) = e^{-\lambda} \lambda^k / k!$.
- Mean = variance = λ .
- Limit of binomial when n large, p small, $np = \lambda$.

15.4 Normal Distribution

- Bell curve, parameters mean μ , variance σ^2 .
- Standard normal $Z = (X - \mu)/\sigma$, mean 0, variance 1.
- Empirical rule: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .
- Symmetric; mean = median = mode.

15.5 Exponential Distribution

- $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$.
- Mean $1/\lambda$; variance $1/\lambda^2$.
- Memoryless property: $P(X>s+t | X>s) = P(X>t)$.

Distribution	Mean	Variance
Binomial	np	$np(1-p)$
Poisson	λ	λ
Uniform $[a,b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential	$1/\lambda$	$1/\lambda^2$
Normal	μ	σ^2

CBT Trap

- Poisson mean = variance = λ ; the exponential (continuous, memoryless) and Poisson (discrete count) describe the same process differently.

16. Statistics

16.1 Measures of Central Tendency

- **Mean** = sum/count (sensitive to outliers).
- **Median** = middle value when sorted (robust to outliers).
- **Mode** = most frequent value.
- Empirical relation: **Mode = 3·Median – 2·Mean**.

CBT Trap

- For **skewed** data the **median** is preferred; mean is pulled toward the tail.

16.2 Measures of Dispersion

- **Range** = max – min.
- **Variance** $\sigma^2 = \Sigma(x-\mu)^2/N$ (population) or $/(n-1)$ (sample).
- **Standard deviation** = $\sqrt{\text{variance}}$.
- **Coefficient of variation** = σ/μ .

CBT Trap

- **Sample variance** divides by **n–1** (Bessel's correction); population divides by **N**.

16.3 Correlation and Regression

- **Pearson correlation** $r \in [-1, 1]$; $r = \text{cov}(X,Y)/(\sigma_X \sigma_Y)$.
- **r = 0** means no **linear** correlation.
- **Regression line**: $y = a + bx$, slope **b** = $\text{cov}(X,Y)/\text{Var}(X)$.
- **r** = $\sqrt{(b_{xy} \cdot b_{yx})}$ (geometric mean of regression coefficients).

CBT Trap

- **Correlation** $\neq$ **causation**; also **r = 0** rules out only **linear**, not all, dependence.

17. Numerical Solutions of Equations

17.1 Non-linear Equations

17.1.1 Bisection Method

- Needs $f(a) \cdot f(b) < 0$ (sign change brackets a root).
- Each step **halves** the interval; **linear** convergence (order **1**).
- Error after n steps $\leq (b-a)/2^n$.
- Always **converges** (guaranteed) but slowly.

17.1.2 Newton-Raphson Method

- Iteration: $x_{(n+1)} = x_n - f(x_n)/f'(x_n)$.
- Quadratic** convergence (order **2**) near a simple root.
- Fails if $f'(x_n) = 0$ or poor initial guess.

17.1.3 Secant / Regula-Falsi

- Secant**: $x_{(n+1)} = x_n - f(x_n)(x_n - x_{(n-1)})/(f(x_n) - f(x_{(n-1)}))$; order ≈ 1.618 .
- Regula-falsi**: like secant but keeps the **bracketing** interval (guaranteed convergence).

Method	Convergence order	Bracketing
Bisection	1 (linear)	yes
Newton-Raphson	2 (quadratic)	no
Secant	1.618	no
Regula-falsi	linear	yes

CBT Trap

- Newton-Raphson** is fastest (order **2**) but needs **f'** and may diverge; **bisection** is slow but **guaranteed**.

17.2 Linear Systems

17.2.1 Gauss Elimination

- Direct method; forward elimination then back substitution; **$O(n^3)$** .

17.2.2 Gauss-Seidel Iterative Method

- Iterative; uses **updated values immediately** (faster than Jacobi).
- Converges if matrix is **diagonally dominant**.

CBT Trap

- Gauss-Seidel** uses new values within the same iteration; **Jacobi** uses only previous-iteration values.

17.3 Convergence and Error

- Absolute error** = $|true - approx|$; **relative error** = $absolute/|true|$.
- Diagonal dominance guarantees iterative convergence.

18. Numerical Integration

18.1 Trapezoidal Rule

- $\int_a^b f(x) dx \approx (h/2)[f_0 + 2(f_1 + \dots + f_{n-1}) + f_n]$, $h = (b-a)/n$.
- Error $O(h^2)$; exact for **linear** (degree 1) polynomials.

18.2 Simpson's 1/3 Rule

- $\int \approx (h/3)[f_0 + 4(\text{odd}) + 2(\text{even}) + f_n]$; needs **even** number of intervals.
- Error $O(h^4)$; exact for **cubic** (degree 3) polynomials.

18.3 Error in Numerical Integration

Rule	Error order	Exact up to degree
Trapezoidal	$O(h^2)$	1
Simpson's 1/3	$O(h^4)$	3
Simpson's 3/8	$O(h^4)$	3

CBT Trap

- **Simpson's 1/3 needs an EVEN number of subintervals** (odd number of points); trapezoidal has no such restriction.
- Simpson's is exact for **cubics** despite using parabolas — a famous bonus accuracy.

19. Numerical Solution of ODEs

19.1 Single-Step Methods

- **Euler's method**: $y_{n+1} = y_n + h \cdot f(x_n, y_n)$; first-order accurate, error **$O(h)$** per step.
- **RK4 (Runge-Kutta 4th order)**: $y_{n+1} = y_n + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)$.
- RK4 slopes: $k_1=f(x,y)$, $k_2=f(x+h/2, y+hk_1/2)$, $k_3=f(x+h/2, y+hk_2/2)$, $k_4=f(x+h, y+hk_3)$.

Method	Order	Slopes used
Euler	1	1
RK2	2	2
RK4	4	4

19.2 Multi-Step Methods

- Use **several previous points** (e.g., **Adams-Bashforth**, **Adams-Moulton**).
- **Predictor-corrector** pairs an explicit predictor with an implicit corrector.

CBT Trap

- **Euler is first-order** (least accurate); **RK4** weights the middle slopes **2,2** and is **fourth-order**.

20. Exam Focus / Practice Checklist

20.1 Propositional Logic Equivalence / Validity

- Test equivalence via **truth tables** or laws; check argument **validity** with rules of inference.
- Remember $p \rightarrow q \equiv \neg p \vee q$ and only the **contrapositive** is equivalent.

20.2 Relation/Function Property Identification

- Classify by **reflexive, symmetric, antisymmetric, transitive**; **RST = equivalence**, **RAT = partial order**.

20.3 Lattice / Hasse Diagram Questions

- Confirm **LUB and GLB** exist for all pairs; identify **0 and 1**, distributive/complemented.

20.4 Counting

- Decide **permutation (order matters) vs combination**; apply **pigeonhole** n/k .

20.5 Recurrence Solving

- Use the **characteristic equation**; remember $(A+Bn)r^n$ for repeated roots.

20.6 Matrix Rank / Determinant / Inverse

- Use **row-echelon** for rank; $A^{-1} = \text{adj}(A)/\det(A)$; $\det(kA) = k^n \det(A)$.

20.7 Eigenvalue Computation and Properties

- **trace** = $\sum \lambda$, **det** = $\prod \lambda$; use **Cayley-Hamilton** for powers/inverse.

20.8 Probability

- **Bayes** for reverse conditioning; memorize **means/variances** of binomial, Poisson, normal, exponential.

20.9 Newton-Raphson / Trapezoidal / Simpson

- **NR**: $x - f/f'$ (order 2); **Trapezoidal** $O(h^2)$; **Simpson** $1/3 O(h^3)$, even intervals.

20.10 Graph Theory

- **Handshaking** $\sum \deg = 2E$; **Euler** $V - E + F = 2$; χ coloring; Euler vs Hamilton conditions.

Final Revision Trick

- Memorize the **distribution mean/variance table**, the **convergence-order table**, and the **eigenvalue trace/det shortcuts** — these recur in nearly every CIL MT mathematics paper.